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Enantioselective Vinylogous Addition of Enones to Allenes Enabled by Synergistic Borane/Palladium Catalysis

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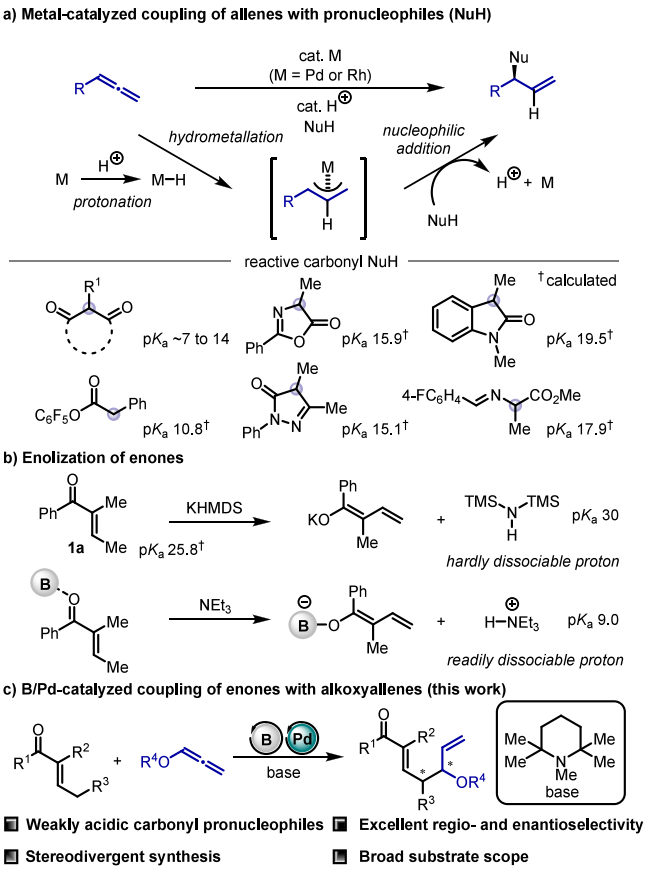
ABSTRACT: Herein, we report a method for enantioselective vinylogous addition of enones to alkoxyallenes enabled by synergistic borane/palladium catalysis. The inductive effect provided by borane coordination to the ketone was essential for closing the gap between the conditions needed for the generation of a dienolate and those needed for initiation of the palladium catalytic cycle by protonation of the metal catalyst. Furthermore, we accomplished the first example of stereodivergent synthesis with chiral borane/transition-metal catalysts.

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Transition-metal-catalyzed coupling of unsaturated hydrocarbons, such as allenes, 1,3-dienes, and alkynes, with a pronucleophile (NuH) is a versatile, atom-economical way to generate allylic compounds, which are widely used for the synthesis of natural products as well as drugs and other bioactive molecules.[1] As shown in Scheme 1a (top panel) with allenes as an example, the first step in these reactions is in situ formation of a M−H species by protonation of the metal catalyst with a proton generated by dissociation of the pronucleophile or a Brønsted acid cocatalyst. The resulting M−H species undergoes a hydrometalation reaction with the unsaturated hydrocarbon to generate a M(π-allyl) complex, which participates in an addition reaction with the nucleophile to deliver the allylic product. Since the pioneering work in this area by the groups of Trost,[2] Yamamoto,[3] and Breit[4] with palladium and rhodium catalysts, highly enantioselective reactions involving various pronucleophiles have been developed.[1,5]

A variety of carbonyl pronucleophiles have been used for this purpose.[2b−e,3a,4e,6] Because both a proton and an enol or enolate nucleophile are required for such reactions, carbonyl pronucleophiles have been confined to compounds with relatively acidic α protons, which can form an enol nucleophile under acidic conditions and/or can dissociate to generate a proton when an acid cocatalyst is not present (Scheme 1a, bottom panel). It can be imagined that if a weakly acidic carbonyl compound was used, access to the corresponding enolate would require a strong base, which would limit the availability of a free proton for protonation of the metal catalyst. To date, the only known reactions of poorly acidic carbonyl compounds, such as arylalkylketones and amides, have involved dienes and nickel catalysis.[7] Unfortunately, in palladium- and rhodium-catalyzed reactions, a proton captured by a strong base has difficulty reacting with the metal. To date, the least acidic carbonyl pronucleophile that has been used for the coupling reaction is 1,3-dimethylindolin-2-one,[2e] the calculated pK_a of which in DMSO is 19.5, as determined by a machine learning approach developed by the Luo group.[8,9]

Scheme 1. Metal-Catalyzed Coupling of Allenes with Carbonyl Pronucleophiles



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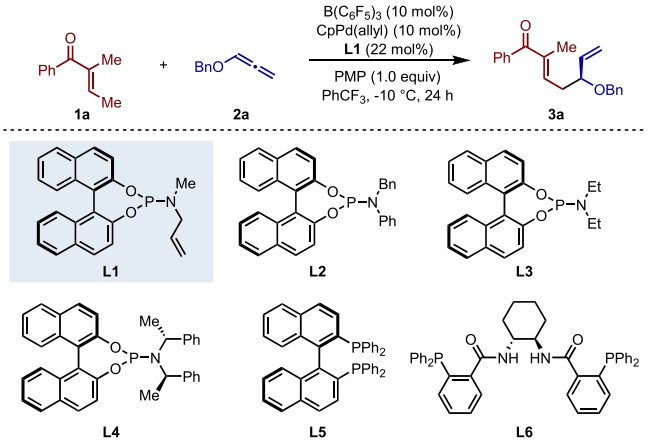
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Enolization of α,β-unsaturated carbonyls has enabled their vinylogous addition reactions with various electrophiles.[10,11] However, vinylogous coupling reactions of α,β-unsaturated carbonyls with allenes, 1,3-dienes, or alkynes have not been reported.[12] This challenge arises from the weak acidity of the γ-protons. For enones (e.g., **1a** with a calculated pK_a of 25.8 in DMSO),[8] enolization can be accomplished with potassium bis(trimethylsilyl)amide (KHMDS).[13] However, bis(trimethylsilyl)amine has a pK_a of 30 in DMSO,[14] so proton dissociation is strongly disfavored (Scheme 1b). Recently, our group developed a method for asymmetric vinylogous Mannich reactions between enones and imines with catalysis by a chiral bicyclic bisborane.[15,16] Coordination of the borane catalyst with the carbonyl oxygen inductively increases the acidity of the γ-proton of the enone, facilitating enolization by a trialkylamine. The steric bulk of the catalyst prevents mutual quenching with the amine base. Such an unquenched borane–amine pair can be described as a frustrated Lewis pair, as defined by Stephan.[17] Because protonated triethylamine has a measured pK_a of 9.0 in DMSO,[14] which is acidic enough to protonate a palladium catalyst,[18] we hypothesized that a synergistic borane/palladium system might be able to promote addition reactions between enones and allenes in the presence of a trialkylamine proton shuttle. Indeed, we herein report that we have developed a method for highly regio- and enantioselective coupling reactions between enones and alkoxyallenes mediated by a synergistic catalyst system and 1,2,2,6,6-pentamethylpiperidine (PMP) as the base (Scheme 1c).

We began by investigating the reaction between enone **1a** and allene **2a** (Table 1). After careful screening of the reaction conditions, we found that target product **3a** could be obtained in 96% yield with 92% enantiomeric excess (ee) by reaction of 0.20 mmol **1a** with 0.30 mmol **2a** in 1.0 mL of $PhCF_3$ at −10 °C for 24 h in the presence of 10 mol % $B(C_6F_5)_3$, 10 mol % CpPd(allyl), 22 mol % ligand **L1**, and 1.0 equiv of PMP (entry 1). The strong Lewis acidity and steric bulk of $B(C_6F_5)_3$ were critical, as indicated by the fact that $B(2,4,6\text{-}F_3C_6H_2)_3$, which is less acidic,[19] and $BF_3{\cdot}OEt_2$, which is less bulky, were inactive (entries 2 and 3). The metal Lewis acids $Sc(OTf)_3$, $Yb(OTf)_3$, and $Mg(OTf)_2$ were also inactive (entries 4–6). Variation of the ligand revealed that phosphoramidite ligands **L2** and **L3** promoted the reaction; **L4**, (*S*)-BINAP (**L5**), and Trost's ligand (**L6**) failed to afford the desired product (entries 7–11). Next, the effect of the base was studied. The bulky base diisopropylethylamine (DIPEA) gave results similar to those obtained with PMP (96% yield and 90% ee, entry 12). The less bulky bases Et_3N and *N*-methylpiperidine gave lower yields and ee values (entries 13 and 14), perhaps because of coordination of the base to the borane or palladium species.[18e] Use of the stronger bases LiHMDS and Barton's base (2-*tert*-butyl-1,1,3,3-tetramethylguanidine) shut down the reaction (entries 15 and 16) probably because of the difficulty of generating the necessary Pd–H species.[20]

Having determined the optimal reaction conditions, we turned our attention to an investigation of the substrate scope. Reactions of **1a** with benzoxyallenes bearing an electron-donating or electron-withdrawing group at the *para* position of the benzene ring gave the desired products (**3b**–**3k**) in 60–96% yields with excellent enantioselectivities (>90% ee) (Table 2). Specifically, halogen atoms (**3c**–**3e**) and methoxy (**3h**), methylthio (**3i**), trifluoromethyl (**3j**), and trifluoromethoxy (**3k**) groups were compatible with the reaction conditions. In addition, *ortho*-fluoro- and *meta*-chloro-substituted benzoxyallenes gave **3l** and **3m**, respectively, with high enantioselectivities. Replacing the benzene ring with a naphthalene ring (**3n**) or a heterocyclic ring (**3o** and **3p**) did not affect the enantioselectivity, but an allene derived from *N*-Boc-piperidine gave **3q** with only 76% ee. An alkoxyallene with a relatively long alkyl chain gave **3r** in 84% yield with 85% ee. To our delight, alkoxyallenes derived from (*S*)-citronellol (**3s**), geraniol (**3t**), (*S*)-perillyl alcohol (**3u**), gemfibrozil (**3v**), and naproxen (**3w**) all gave the corresponding products in high yields with high diastereo- or enantioselectivities. Moreover, a gram-scale reaction of **1a** with **2h** gave **3h** in 87% yield and 92% ee (see the Supporting Information). Unfortunately, when phenylallene, benzylallene, or 1-benzyloxy-1-methylallene was reacted with **1a** under the standard reaction conditions, no product formation was observed (see the Supporting Information for details on limitations). Notably, alkoxyallenes are known to be more

Table 1. Optimization of Reaction Conditions[a]

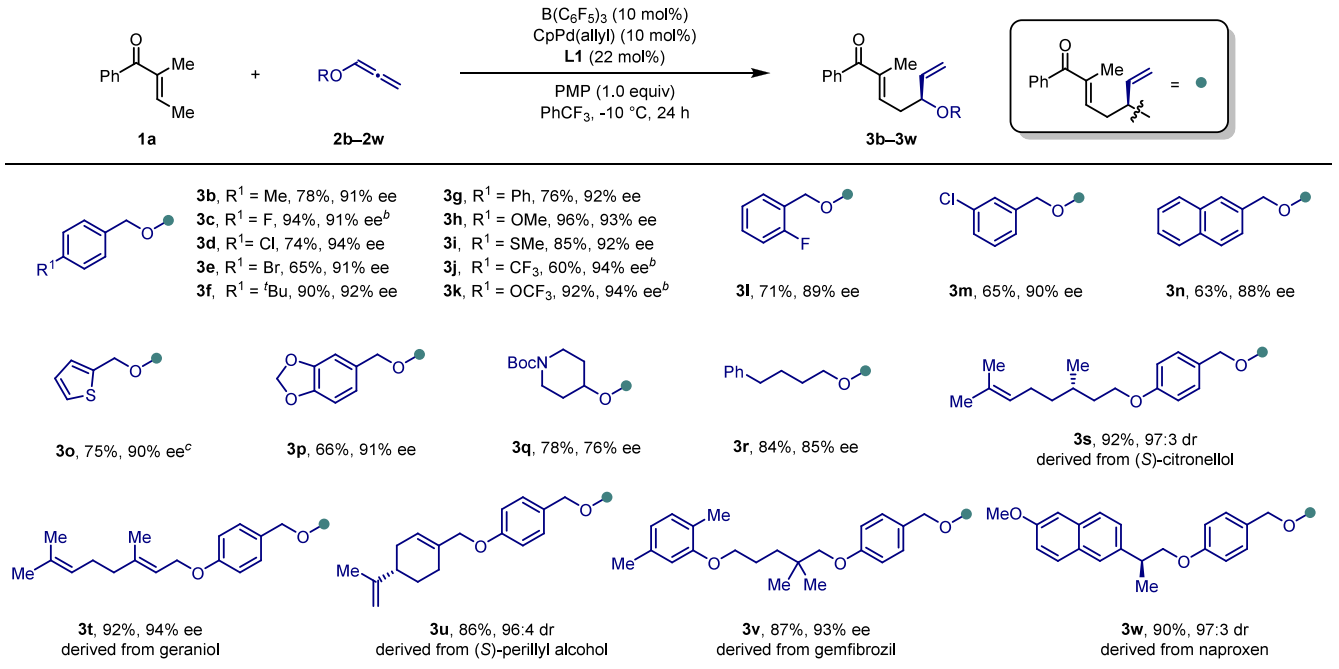


entry	deviation from standard conditions	yield (%)	ee (%)
1	none	96[b]	92
2	$B(2,4,6\text{-}F_3C_6H_2)_3$ instead of $B(C_6F_5)_3$	trace	ND
3	$BF_3{\bullet}OEt_2$ instead of $B(C_6F_5)_3$	trace	ND
4	$Sc(OTf)_3$ instead of $B(C_6F_5)_3$	trace	ND
5	$Yb(OTf)_3$ instead of $B(C_6F_5)_3$	NR	ND
6	$Mg(OTf)_2$ instead of $B(C_6F_5)_3$	NR	ND
7	**L2** instead of **L1**	31	80
8	**L3** instead of **L1**	90	90
9	**L4** instead of **L1**	trace	ND
10	**L5** instead of **L1**	NR	ND
11	**L6** instead of **L1**	NR	ND
12	DIPEA as the base	96	90
13[c]	Et_3N as the base	58	68
14[c]	*N*-methylpiperidine as the base	65	70
15	LiHMDS as the base	NR	ND
16	Barton's base as the base	NR	ND

[a]Unless otherwise specified, all reactions were performed with 0.20 mmol **1a**, 0.30 mmol **2a**, 10 mol % $B(C_6F_5)_3$, 10 mol % CpPd(allyl), 22 mol % ligand, and 0.20 mmol base in 1.0 mL of $PhCF_3$ at −10 °C for 24 h under N_2. NMR yields are reported and were determined by using 4-dimethylaminopyridine as an internal standard, and ee values were determined by HPLC analysis on a Daicel CHIRALCEL OJ-H column. NR, no reaction; ND, not determined. [b]Isolated yield. [c]Reaction at room temperature.

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Table 2. Scope of Alkoxyallenes[a]



[a]Unless otherwise specified, all reactions were performed with 0.20 mmol **1a**, 0.30 mmol alkoxyallene (**2b**−**2w**), 10 mol % $B(C_6F_5)_3$, 10 mol % CpPd(allyl), 22 mol % **L1**, and 0.20 mmol PMP in 1.0 mL of $PhCF_3$ at −10 °C for 24 h under N_2. Isolated yields are provided. Abbreviation: dr, diastereomer ratio. [b]Reaction at 0 °C. [c]Reaction at −15 °C.

reactive than alkyl- or aryl-substituted allenes in Pd-catalyzed hydrofunctionalization reactions of allenes.[1d]

Next, we investigated the scope with respect to the enone by using benzyloxyallene (**2a**) as the coupling partner (Table 3). Aryl ketones with various substituents on the benzene ring attached to the carbonyl group were tolerated, giving corresponding products **4b**−**4m**. Functionality such as halogen atoms (**4d**−**4f**, **4m**) and ester (**4h**), vinyl (**4i**), and alkynyl (**4j**) groups were retained. Moreover, the benzene ring could be replaced by a 2-naphthyl (**4n**), 5-benzofuranyl (**4o**), 5-benzothiophenyl (**4p**), or 2-thienyl (**4q**) group. However, little of the desired product was obtained when the aryl group was replaced with an alkynyl or alkyl group under the standard conditions (data not shown). Interestingly, however, we found that these substrates could be converted if $B(C_6F_5)_3$ was replaced with one of the chiral spirobicyclic bisborane catalysts developed in our group (**CAT1** or **CAT2**).[21] With these catalysts, reactions of alkynyl-substituted enones gave the target products (**4r**−**4t**) with high enantioselectivities, and a cyclohexyl-substituted ketone gave **4u** in 69% yield with 82% ee. The activity of **CAT1** and **CAT2** in these reactions is similar, with the difference lying in their enantioselectivity. A matched/mismatched effect also impacts the enantioselectivity. For example, the use of *ent*-**CAT2** with **L1** in the reaction of **1r** afforded **4r** in 71% yield and 73% ee. Notably, even though ketone **1u** has an additional enolization site on the cyclohexyl group, no side product resulting from enolate addition at this site was detected. However, replacing the cyclohexyl group in **1u** with methyl or *t*-butyl groups results in no reaction or the formation of only trace amounts of products.

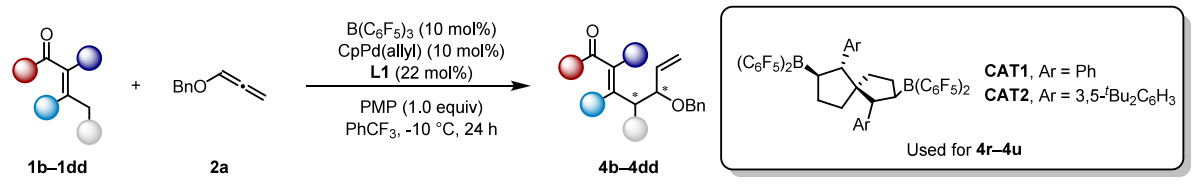
Subsequently, we tested enones with various α and β substituents. When the α substituent was ethyl, *n*-butyl, benzyl, cyclopropyl, or alkynyl, corresponding products **4v**−**4z**, respectively, were obtained in 79−90% yields with 84−92% ee. However, the ee dropped to 76% when the ketone had an α-hydrogen atom (**4aa**). Interestingly, β-disubstituted enone **1bb** was also reactive, giving **4bb** in 42% yield with 90% ee, despite the increased steric hindrance. When we replaced the β-methyl group with a benzyl or *n*-butyl group, products **4cc** and **4dd**, which have adjacent stereogenic centers, were formed in good yields but with poor diastereoselectivity. Notably, no byproducts generated by reactions at the α position of the enones were detected in any of the above-mentioned reactions in Tables 2 and 3, perhaps because of the steric bulk of the borane catalyst. The absolute configuration of **4h** was determined by X-ray single-crystal analysis (CCDC 2401879), whereas the configurations of the other products in Tables 2 and 3 were tentatively assigned by analogy.

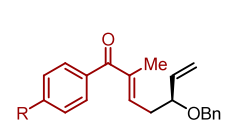
Inspired by the fact that the combination of a transition-metal catalyst with an organocatalyst or a Lewis acid catalyst has been successfully used for many stereodivergent reactions,[22] we wondered whether the catalytic system described herein could also be used for stereodivergent synthesis. Satisfyingly, we found that by varying the stereochemistry of chiral spirobicyclic bisborane catalyst **CAT2**, we were able to control the stereochemistry of the γ stereogenic center of the product (**4ee**) generated by reaction of alkynyl-substituted enone **1ee** with **2a** and that by varying the ligand for the palladium catalyst, we could control the stereochemistry of the δ stereogenic center (Scheme 2).[23−25] In this way, we were able to achieve diastereoselective formation of all four stereoisomers with high enantioselectivities.

On the basis of literature reports,[1,18] we propose that these synergistic catalytic reactions proceed via the mechanism shown in Scheme 3. The catalytic cycle is initiated by coordination of the borane catalyst to the carbonyl group of

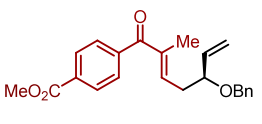
https://doi.org/10.1021/jacs.4c16214
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Table 3. Scope of α,β-Unsaturated Ketones[a]

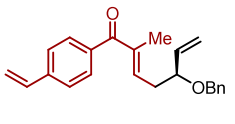




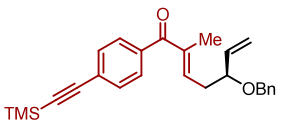
4b, R = Me, 91%, 90% ee
4c, R = Ph, 84%, 91% ee[b]
4d, R = F, 62%, 93% ee
4e, R = Cl, 86%, 92% ee[c]
4f, R = Br, 58%, 90% ee[c]
4g, R = TMS, 84%, 88% ee



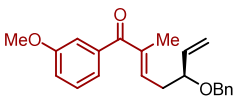
4h, 76%, 93% ee



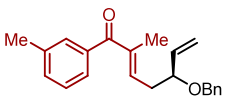
4i, 86%, 91% ee



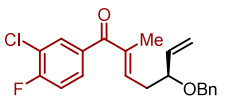
4j, 53%, 94% ee[d]



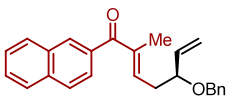
4k, 82%, 90% ee[c]



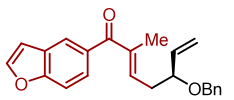
4l, 78%, 90% ee[b]



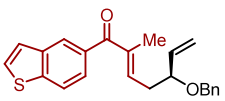
4m, 48%, 91% ee



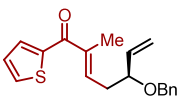
4n, 80%, 88% ee



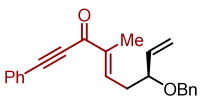
4o, 84%, 90% ee



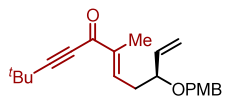
4p, 84%, 86% ee



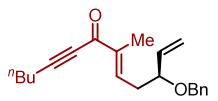
4q, 70%, 93% ee



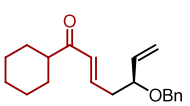
4r, 80%, 92% ee[e]



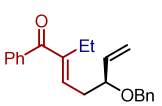
4s, 70%, 89% ee[f]



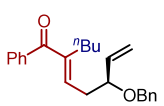
4t, 56%, 90% ee[e]



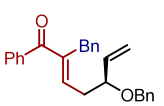
4u, 69%, 82% ee[g]



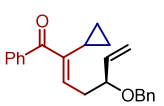
4v, 90%, 92% ee



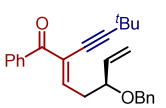
4w, 79%, 91% ee



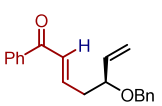
4x, 90%, 87% ee



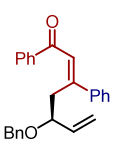
4y, 90%, 86% ee



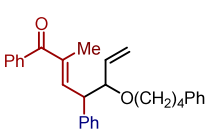
4z, 80%, 84% ee



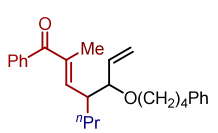
4aa, 77%, 76% ee[h]



4bb, 42%, 90% ee



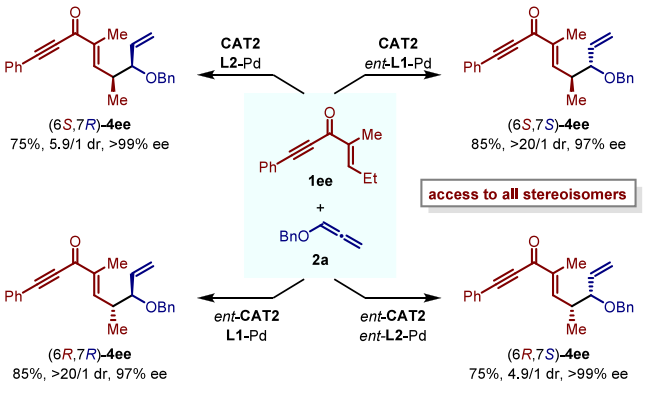
4cc, 87%, 1.3/1 dr
major, 68% ee; minor, 90% ee[i]



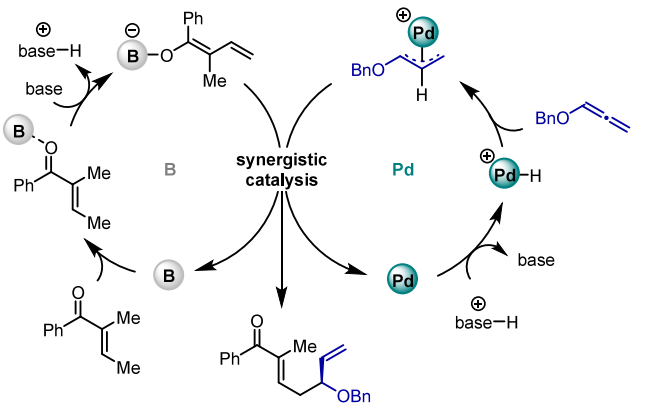
4dd, 71%, 2.4/1 dr
major, 63% ee; minor, 90% ee[i]

[a]Unless otherwise specified, all reactions were performed with 0.20 mmol enone (**1b**–**1dd**), 0.30 mmol **2a**, 10 mol % $B(C_6F_5)_3$, 10 mol % CpPd(allyl), 22 mol % **L1**, and 0.20 mmol PMP in 1.0 mL of $PhCF_3$ at −10 °C for 24 h under N_2. Isolated yields are provided. [b]Reaction at −20 °C. [c]0.5 mL of $PhCF_3$ was used. [d]20 mol % $B(C_6F_5)_3$ was used. [e]5 mol % **CAT2** was used at −20 °C. [f]0.30 mmol **2h** and 5 mol % *ent*-**CAT1** were used at −20 °C. PMB = *p*-methoxybenzyl. [g]5 mol % **CAT1** was used at −10 °C. [h]Reaction at 0 °C. [i]0.30 mmol **2r** and 20 mol % $B(C_6F_5)_3$ were used at 0 °C.

Scheme 2. Stereodivergent Synthesis



Scheme 3. Proposed Catalytic Cycle



the enone, which inductively activates the γ-C−H bond. Subsequent deprotonation with the base generates a dienolate. The protonated base then enters the palladium cycle to effect protonation of Pd(0) to give Pd(II)-H. Hydropalladation of the allene with Pd(II)-H generates a Pd(π-allyl) complex, which react with the dienolate to give the final product, with regeneration of both the borane and the palladium catalysts.

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In summary, we have developed a method for regio- and enantioselective coupling of enones with alkoxyallenes with synergistic borane/palladium catalysis. The strong inductive effect of coordination of the borane catalyst to the carbonyl group of the enone enabled trialkylamine-promoted enolization, a process that is otherwise impossible because of the high pK_a values of these compounds. The resulting protonated trialkylamine could protonate the palladium catalyst to initiate the palladium catalytic cycle. In this way, the pK_a limit for carbonyl pronucleophiles in palladium-catalyzed coupling reactions with unsaturated hydrocarbons has been increased to approximately 26. Furthermore, by combining chiral spirobicyclic bisborane catalysts with chiral palladium catalysts, we accomplished stereodivergent synthesis. We believe that such synergistic borane/transition-metal catalytic systems could push the boundaries of well-established catalytic systems with transition metals, thereby leading to the discovery of new reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c16214.

Experimental details, compound characterization data, and NMR spectra (PDF)

Accession Codes

Deposition Numbers 2401638 and 2401879 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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(23) The absolute configuration of (6*R*,7*R*)-**4ee** was determined by X-ray single-crystal diffraction of its derivative (CCDC 2401638).
(24) The slight variation in diastereoselectivity observed during the formation of (6*S*,7*R*)-**4ee** and (6*R*,7*S*)-**4ee** may be attributed to impurities in the catalysts.
(25) The use of **CAT2** and *ent*-**L2**-Pd gave (6*S*,7*S*)-**4ee** in 80% yield, 8.4/1 dr, and 94% ee.

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